



About Us

The InsectAI COST Action supports insect monitoring and conservation at national and continental scales addressing widespread insect declines. It unites researchers and stakeholders in image-based insect AI technologies to shape research agendas, enhance research capacity across Europe, and promote innovation.

Given the global decline in insect populations, understanding species trends and drivers is critical for effective conservation. Efficient and standardized insect monitoring is essential to inform sustainable biodiversity and economic strategies. The EU Biodiversity Strategy 2030 and Nature Restoration Law highlight the need for high-resolution, standardized data, which InsectAI can help provide.

The Action facilitates workshops, conferences, and hackathons across four Working Groups, focusing on societal needs, image collection, standardization, and data analysis to deliver actionable insights and contribute to EU policies.



Contact:

info@insectai.eu
www.insectai.eu

Follow us and join us!



<https://linktr.ee/insectai>

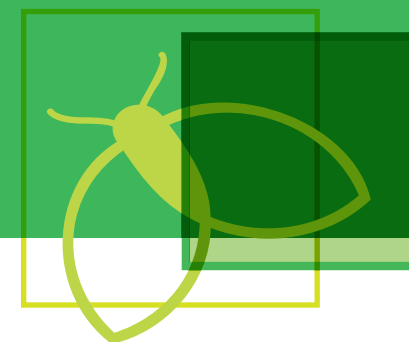
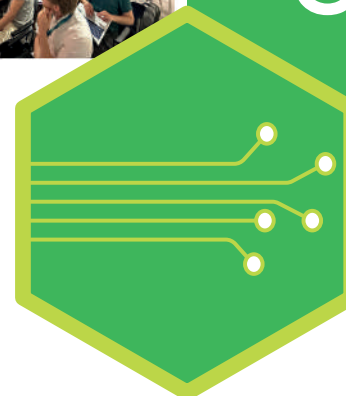
This publication is based upon work from COST Action

CA22129 - Insect AI
Using Image-based AI for Insect
Monitoring & Conservation
supported by COST

COST (European Cooperation in Science and Technology) is a funding agency for research and innovation networks. Our Actions help connect research initiatives across Europe and enable scientists to grow their ideas by sharing them with their peers. This boosts their research, career and innovation.
www.cost.eu



Using Image-based AI for Insect Monitoring & Conservation



InsectAI



Working Group 1 Societal Needs

Engaging diverse stakeholders through participatory methods to ensure that image-based insect AI technologies benefit a wide range of end-users.

- Prioritizing Societal Needs: Identifying critical research questions and aligning technologies with policy and societal requirements.
- Stakeholder Involvement: Engaging researchers, citizen scientists, and policymakers to guide the development and application of new tools.
- Effective Dissemination: Creating resources to ensure smooth knowledge transfer and maximize technology adoption.



Working Group 2 Image Collection

Supporting the development of image collection

technologies and amplifying their impact on AI-based analysis for large-scale insect monitoring, both in nature and under laboratory conditions.

- Collaboratively developing best practices guides across working groups, building community capacity, and promoting the adoption of new monitoring methods.
- Facilitating collaborations between hardware developers and users, running workshops and training schools, and engaging with stakeholders to meet specific monitoring needs.
- Promoting citizen-science innovation in automated monitoring.



Working Group 3 Image Processing

Advancing image processing for insects to ensure high-quality, standardized tools for consistent species identification.

- Key objectives include standardising data storage, aligning workflows, and enhancing insect detection and classification models.
- Activities will focus on collecting annotated data, improving algorithm robustness, developing metadata standards, and defining infrastructure for sharing models.
- Knowledge sharing activities will include workshops to develop guidelines and hackathons to develop novel algorithms.

